

1. Use Part 1 of the Fundamental Theorem of Calculus to find the **derivative** of $F(x) = \int_0^{\sin x} \frac{1}{t^3 + 1} dt$.

2. Evaluate the following integral by u -substitution. Make sure to indicate the u -function.

$$\int \frac{x^3}{(x^4 - 2)^3} dx$$